

MUHAMMAD FAQIH

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SUMMARY

AI/ML Engineer with production experience deploying and optimizing large language models, building end-to-end ML pipelines, and shipping real AI systems at government scale. Experienced in LLM serving via vLLM, model quantization, multimodal inference, and integrating AI capabilities into high-concurrency backend services built with FastAPI and PostgreSQL. I work across the full ML lifecycle, from research and experimentation to production deployment and monitoring.

EXPERIENCE

AI Engineer

Pasuruan Regency Government – Department of Communications & Information Technology January 2026 - Present, Pasuruan, Indonesia

- Independently engineered the full lifecycle of Kapas AI (ai.pasuruankab.go.id), a government-grade multimodal AI platform, covering raw GPU server provisioning, Linux administration, model deployment, and frontend delivery.
- Deployed and optimized Qwen3.5-9B via vLLM with BitsAndBytes half-precision quantization, maintaining 95% GPU memory utilization. Designed a custom sliding window memory with background LLM-based auto-summarization to maintain long-term context within a 8192-token window.
- Developed high-concurrency microservices in FastAPI (Python 3.12) with async/await and real-time SSE streaming. Built an OpenAI-compatible API gateway with custom auth middleware, sliding window rate-limiting, and asyncio concurrency control.
- Designed complex PostgreSQL relational schemas via SQLAlchemy 2.x ORM with UUID indexing, many-to-many associations, and automated migrations via Alembic.
- Built automated pipelines for multi-format document extraction (PDF, DOCX, TXT) and real-time image preprocessing, feeding into a production RAG system using Qdrant, BGE-M3 hybrid embeddings, and a BGE reranker for two-stage retrieval to ground Kapas AI's responses in verified government documents.
- Created an automated forecasting service across 12 regional tax types and 24 districts, implementing SARIMA and Damped Holt's Linear Trend with automatic model selection; evaluated using sMAPE and MAE to ensure stability on volatile fiscal data for regional budget planning.
- Architected an extensible scraping framework using Abstract Base Classes to aggregate 300–500 posts/cycle from Instagram, TikTok, and Facebook via Apify SDKs; integrated Google Gemini and IndoBERT in a dual-model pipeline for batch sentiment classification and narrative topic modeling.
- Built a fault-tolerant background worker system via APScheduler (~95% job success rate) automating daily scraping and AI analysis cycles with independent database session management.

Operational Excellence (Internship)

Ajinomoto Indonesia

July 2025—August 2025, Mojokerto, Indonesia

- Assisted in collecting, organizing, and consolidating operational data for internal reporting
- Reduced manual and paper-based reporting processes by approximately 60% through digital transformation
- Prepared structured reports and summaries to support management review and decision-making

RESEARCH & PUBLICATION

- Faqih, M., Rahman, R. A., & Holle, K. F. H. (2026). Efficient and robust crosswalk segmentation under adverse weather using ConvNeXt-enhanced DeepLabv3. Jurnal Ilmu Komputer dan Informasi, 19(2), 179–195. <http://dx.doi.org/10.21609/jiki.v19i2.1757>
- Faqih, M., Aziz, O. Q., & Hanani, A. (2026). A ConvNeXt-based deep learning framework for endoscopic image classification in GERD and polyp detection. Jurnal Ilmu Komputer dan Informasi, 19(2), 157–167. <http://dx.doi.org/10.21609/jiki.v19i2.1702>

PROJECTS

End-to-End MLOps Platform for Endoscopic Diagnostics: GERD & Polyp Detection

Personal Project • <https://github.com/mxfaqih/gerdpolyp-mlops>

- Transformed research project (12 experimental scenarios on endoscopy classification using modern CNN Architecture) into production-ready MLOps system, bridging academic research and industry deployment standards
- Built comprehensive ML infrastructure: migrated from notebook-based prototypes to modularized codebase with DVC versioning, MLflow tracking, automated reproducibility, and full experiment traceability
- Deployed containerized inference API with FastAPI, achieving <600ms latency with optimized model loading and preprocessing production-ready for clinical integration

EDUCATION

Bachelor of Computer Science

State Islamic University of Maulana Malik Ibrahim Malang (August 2022 – December 2025)

GPA: 3.62/4.0

- Relevant Courses: Artificial Intelligence, Computer Vision, Natural Language Processing, Distributed Systems, Data Science, Numerical Methods, Database, Statistics
- Organization: Vice President, Data Science Enthusiast Community (Department - Affiliated Student Organization) | September 2024 – July 2025

SKILLS

Programming: Python, Java, HTML, CSS, PHP, Javascript, MySQL, PostgreSQL, MongoDB, Docker, Git, GitHub Action, AWS
AI & Machine Learning: TensorFlow, PyTorch, Keras, Hugging Face, OpenCV, FastAPI, Flask, Data Version Control, MLFlow, Qdrant
Language: English (Profesional Working Proficiency), Japanese (Professional Working Proficiency, N2)

CERTIFICATION

- Associate Data Scientist – National (Ministry of Communication and Information Technology of the Republic of Indonesia) – February 2026
- Machine Learning Operations (MLOps) with Cloudeka – (Dicoding Indonesia & Lintasarta) – September 2024